higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N5

28 NOVEMBER 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1 $T_A = 2T_P + T_S$
$= 2(44) + 22$ ✓
$= 110$ ✓

No	Condition	Gear A	Gear P	Gear S	Arm L
1	Fix arm L and rotate gear A +1 rev	+1	$\frac{T_A}{T_P} = \frac{110}{44} = 2,5$	$\frac{T_A}{T_P} x \frac{T_P}{T_S} = \frac{110}{44} x \frac{44}{22} = -5$	0
2	Multiply by x and add y	$x + y$	$2x + y$	$-5x + y$	y
3	NS = 290 NA = 0 NL = ?	0		290	N_L?
4	NL = 18 NS = 290 NA = ?	NA = ?		290	18

(2)

1.2 $x + y = 0$
$x = -y$✓
$-5x + y = 290$
$-5(-y) + y = 290$✓
$6y = 290$
$y = \frac{290}{6}$
$y = 48,333$✓
$N_L = y$
$N_L = 48,333 r/min$✓
$P_{in} = \frac{2\pi N T_{in}}{60}$
$T_{in} = \frac{60 x P_{in}}{2\pi N}$
$T_{in} = \frac{60x4000}{2\pi x290}$✓
$T_{in} = 131,714 N.m$✓
$(T_{in} x N_{in}) + (T_{out} x N_{out}) + ((T_{hold} x N_{hold}) = 0$
$(131,714x290) + (T_{out} x 48,333) + 0 = 0$✓
$T_{out} = \frac{-38197,06}{48,333}$
$T_{out} = -790,284\ N.m$✓
$T_{out} = -790,284\ x \frac{86}{100}$✓
$T_{out} = -679,644\ N.m$✓

(10)

1.3

$$y = 18$$
$$-5x + y = 290$$
$$-5x + 18 = 290$$
$$-5x = 272$$
$$x = \frac{272}{-5} \checkmark$$
$$x = -54{,}4 \checkmark$$
$$N_A = x + y$$
$$N_A = -54{,}4 \ + 18 \checkmark$$
$$N_A = -36{,}4r/min \checkmark$$

(4)
[16]

QUESTION 2

2.1 2.1.1

$$\frac{T_G}{T_F} = \frac{T_k}{T_H}$$
$$\frac{10}{1} = \frac{T_k^2}{T_H^2}$$
$$10x(18)^2 = T_k^2 \checkmark$$
$$T_k = \sqrt{4000}$$
$$T_k = 56{,}92$$
$$Say = 57 \checkmark$$
$$\frac{m}{2}(T_F + T_G) = \frac{m}{2}(T_H + T_G)$$
$$\frac{4}{2}(T_F + T_G) = \frac{6}{2}(18 + 57)$$
$$T_F + T_G = \frac{225}{2}$$
$$T_F + T_G = 112{,}5 \checkmark \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$
$$\frac{10}{1} = \frac{T_G^2}{T_F^2}$$
$$\frac{T_G}{T_F} = \sqrt{10} \checkmark$$
$$T_G = 3{,}16T_F \checkmark \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Sub (2) in (1)

$$T_F + T_G = 112{,}5$$
$$T_F + 3{,}16T_F = 112{,}5 \checkmark$$
$$4{,}16T_F = 112{,}5$$
$$T_F = 27{,}04 \checkmark$$

Say ^{27}

$$T_G = 3{,}16T_F$$
$$= 3{,}16x27 \checkmark$$
$$T_G = 85{,}32$$

Say $^{85}\checkmark$

(9)

2.1.2 $\frac{N_C}{N_D} = \frac{T_G x T_K}{T_F x T_H}$ ✓

$\frac{N_C}{N_D} = \frac{85x57}{27x18}$ ✓

$= 9{,}97:1$ ✓ (3)

2.2
- Increases the leverage of the engine over the driving shaft
- Generates power to carry the load more effectively from rest
- Provides a medium through which the engine can be disengaged
- Provides a reverse gear that allows the engine to continue to run in the same direction (4)

[16]

QUESTION 3

3.1
- Lower installation costs
- Installation is easier
- Cheaper maintenance
- Does not require stronger bearings to support the head pulley shaft (4)

3.2 3.2.1 $v = \frac{\pi(D + t)N}{60}$

$= \frac{\pi(0{,}7 + 0{,}004)750}{60}$ ✓

$v = 27{,}65 m/s$ ✓ (2)

3.2.2 $m = wxtxlx\rho$

$= 0{,}12x0{,}004x1x1{,}2x10^3$ ✓

$m = 0{,}576 kg/m$ ✓ (2)

3.2.3 $T_c = mv^2$

$= 0{,}576x(27{,}65)^2$ ✓

$T_c = 440{,}365N$ ✓ (2)

3.2.4 $T_1 = wxtxft$

$= 120x4x2,5$

$T_1 = 1200N$ ✓

$$\frac{T_1 - T_c}{T_2 - T_c} = e^{\mu\theta}$$

$$\frac{1200 - 440,365}{T_2 - 440,365} = 2,718^{0,25x\frac{160}{57,3}}$$ ✓

$$\frac{759,635}{T_2 - 440,365} = 2,01$$ ✓

$2,01T_2 - 885,085 = 759,635$

$T_2 = 818,313N$ ✓ (4)

3.2.5 $P = (T_1 - T_2)vx\eta$

$= (1200 - 818,313)27,65x0,9$ ✓

$P = 9,5kW$ ✓ (2)

[16]

QUESTION 4

4.1
- Malfunctioning of chain and belt tensioners causing chain and belt slackness resulting in loss in power
- Overloading of buckets
- Uneven spacing of buckets on chain or belt (3)

4.2 4.2.1 $v = \sqrt{gcos\theta r}$

$v^2 = gcos\theta r$

$$r = \frac{v^2}{gcos\theta}$$

$$v = \frac{90}{60} = 1,5m/s$$ ✓

$$r = \frac{1,5^2}{9,81xcos45}$$ ✓

$r = 0,324m$ ✓

$d = 0,324x2$

$d = 0,648m$

$d = 648mm$ ✓ (4)

4.2.2 $L = 2\pi r + 2C$

$L = 2\pi x0,324 + 2x95$ ✓

$L = 192,036m$ ✓ (2)

4.2.3

$$B_{/s} = \frac{v}{S_{bu}}$$
$$B_{/s} = \frac{1,5}{0,72} \checkmark$$
$$B_{/s} = 2,083$$
$$say = 2buckets/sec \checkmark$$

(2)

4.2.4

$$m_{/b} = \frac{m}{b_{/s}}$$
$$m_{/b} = \frac{1500x10^3}{7x3600x2} \checkmark\checkmark$$
$$m_{/b} = 29,76kg_{/bucket} \checkmark$$

(3)

4.2.5

$$P_g = mgh$$
$$h = \frac{total\ length}{2}$$
$$h = \frac{192,036}{2}$$
$$h = 96,018m$$

$$P_g = mgh$$
$$= 59,52x9,81x96,018 \checkmark$$
$$P_g = 56064,065W \checkmark$$

$$\eta = \frac{P_{out}}{P_{in}}$$
$$P_{in} = \frac{P_{out}}{\eta}$$
$$P_{in} = \frac{56064,065}{0,82} \checkmark$$
$$P_{in} = 68370,811W$$
$$P_{in} = 68,371kW \checkmark$$

(4)
[18]

QUESTION 5

5.1 $P_{nett} = P_{motor} x \eta$

$= 70x0,75$

$P_{nett} = 52,5kW$ ✓

$P_{nett} = Effective\ pull\ x\ v$

$Effective\ pull = \frac{P_{nett}}{v}$

$Effective\ pull = \frac{52,5\ x\ 10^3 x3600}{4500}$ ✓

$Effective\ pull = 42000N$ ✓

$But\ effective\ pull = pull\ against\ gravity + total\ rolling\ resistance$

$Let\ number\ of\ cars/side = X$

$Pull\ against\ gravity = out-of-balance\ load x g x number\ of\ cars/side x gradient$

$Pull\ against\ gravity = 680x9,81xXx\frac{1}{10}$ ✓

$Pull\ against\ gravity = 667,08X$ ✓

$Total\ rolling\ resistance = Total\ moving\ mass(in\ ton) x\ resistance/ton$

$Total\ R_R = (mass\ loaded\ cars + mass\ empty\ cars + mass\ rope) x\ resistance/ton$

$= [(0,95xX) + (0,27xX) + (2x950x0,0025]x270$ ✓

$= (1,22X + 4,75)270$ ✓

$= 329,4X + 1282,5$ ✓

$Total\ effective\ pull = pull\ against\ gravity + total\ rolling\ resistance$

$42000 = 667,08X + 329,4X + 1282,5$

$42000 = 996,48X + 1282,5$ ✓

$996,48X = 40717,5$

$X = \frac{40717,5}{996,48}$ ✓

$X = 40,86 \quad Say\ 41$

$\therefore No\ of\ trucks/side = 41$ ✓

(11)

5.2 $$No.of\ cars/side = \frac{Length\ of\ haulage}{Spacing}$$

$$Spacing = \frac{Length\ of\ haulage}{No.of\ cars/side}$$

$$= \frac{950}{41} \quad \checkmark$$

$$Spacing = 23{,}17m \quad \checkmark$$

$$Spacing = \frac{rope\ speed(m/min)}{No.of\ trucks/min}$$

$$No.of\ trucks/min = \frac{rope\ speed(m/min)}{Spacing}$$

$$No.of\ \frac{trucks}{min} = \frac{4500}{23{,}17x60} \quad \checkmark$$

$$No.of\ \frac{trucks}{min} = 3{,}237 \quad \checkmark$$

$$No.of\ trucks/min = \frac{ton/shift}{60\ x\ hours\ x\ ton/car}$$

$$Ton/shift = No.of\ trucks/min\ x\ 60\ x\ hours\ x\ ton/car \quad \checkmark$$

$$Ton/shift = 3{,}237x60x7x0{,}68 \quad \checkmark$$

$$Ton/shift = 924{,}487ton \quad \checkmark$$

(7)
[18]

QUESTION 6

6.1 $$T_E = m_1 x g x \mu x \eta$$

$$= 140x10^3x9{,}81x0{,}8 \quad \checkmark$$

$$T_E = 329{,}616kN \quad \checkmark$$

(2)

6.2 $$F_R = (M_L x R_L) + (M_T x R_T)$$

$$= (140x150) + (600x110) \quad \checkmark$$

$$F_R = 87kN \quad \checkmark$$

$$F_g = (M_L + M_T)x10^3xgx\frac{1}{140}$$

$$F_g = (140 + 600)x10^3xgx\frac{1}{140} \quad \checkmark$$

$$F_g = 51{,}853kN \quad \checkmark$$

$$T_E = F_R + F_g + F_a$$

$$F_a = T_E - F_R - F_g$$

$$= 329{,}616 - 87 - 51{,}853$$

$$F_a = 190{,}763kN \quad \checkmark$$

(5)

6.3 $T_E = 329,616x\eta$

$= 329,616x0,72$ ✓

$T_E = 237,324kN$ ✓

$F_a = T_E - F_R - F_g$

$= 237,324 - 87 - 51,853$ ✓

$F_a = 98,471kN$ ✓

$F_a = m.a$

$a = \frac{F_a}{m}$

$= \frac{98,471x10^3}{(140 + 600)x10^3}$

$a = 0,133m/s^2$ ✓

$v = \frac{60x10^3}{3600}$

$= 16,667m/s$

$v = u + at$

$t = \frac{v - u}{a}$

$t = \frac{16,667 - 0}{0,133}$ ✓

$t = 125,313\ sec$ ✓ (7)

6.4 $v^2 = u^2 + 2.a.s$

$s = \frac{v^2 - u^2}{2.a}$

$= \frac{16,667^2 - 0^2}{2x0,133}$ ✓

$s = 1044,32m$ ✓ (2)

[16]

TOTAL: 100

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